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LABORATORY RECORD

EXP NO.: 1 SETTING UP HYPERLEDGER FABRIC DEVELOPMENT ENVIRONMENT

AIM

To install and configure the Hyperledger Fabric development environment on a Linux system using Docker, Go, and Fabric binaries, create a blockchain test network, and establish a communication channel between peer organizations.

PROCEDURE

Step 1: Update the System

```
sudo apt update && sudo apt upgrade -y
```

Step 2: Install Required Packages

```
sudo apt install -y git curl jq docker-compose
git --version
curl --version
docker-compose --version
```

Step 3: Verify Docker Installation

```
docker --version
sudo systemctl start docker
sudo systemctl enable docker
sudo usermod -aG docker $USER
newgrp docker
docker run hello-world
```

Step 4: Verify Go Installation

```
go version
```

Step 5: Create a Workspace

```
mkdir -p ~/fabric-dev
cd ~/fabric-dev
pwd
```

Step 6: Download Hyperledger Fabric Samples

```
curl -sSL https://bit.ly/2ysbOFE | bash -s
```

Step 7: Navigate to the Fabric Samples Directory

```
cd fabric-samples
```

Step 8: Configure Environment Variables

```
echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin' >> ~/.zshrc
source ~/.zshrc
```

Step 9: Verify Hyperledger Fabric Installation

```
peer version
orderer version
```

Step 10: Start the Hyperledger Fabric Test Network

```
cd ~/fabric-dev/fabric-samples/test-network
./network.sh up
```

Step 11: Verify Running Containers

```
docker ps
```

Step 12: Create a Channel

```
./network.sh createChannel -c mychannel
```

Step 13: Stop the Network

```
./network.sh down
```

PROGRAM

Fabric Environment Setup & Network Automation Script (setup_fabric_env.sh):

```
#!/bin/bash
# Script for Setting Up Hyperledger Fabric Development Environment & Network
Initialization

set -e

# 1. System Update and Dependency Verification
sudo apt update -y
sudo apt install -y git curl jq docker-compose

# 2. Workspace Setup and Fabric Installation
mkdir -p ~/fabric-dev
cd ~/fabric-dev
curl -sSL https://bit.ly/2ysbOFE | bash -s

# 3. Path Configuration
export  PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin
echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin' >> ~/.zshrc

# 4. Binary Version Validation
peer version
orderer version

# 5. Network Lifecycle Execution
cd  ~/fabric-dev/fabric-samples/test-network
./network.sh up
docker ps
./network.sh createChannel -c mychannel
./network.sh down

echo "Hyperledger Fabric Development Environment Configured Successfully."
```

OUTPUT

```
user@ubuntu:~/fabric-dev$ git --version && docker --version && go version
git version 2.43.0
Docker version 24.0.7, build af150ea
go version go1.22.0 linux/amd64

user@ubuntu:~/fabric-dev$ peer version && orderer version
peer:
 Version: v2.5.16
 Commit SHA: d2a013f
 Go version: go1.20.12
 OS/Arch: linux/amd64

orderer:
 Version: v2.5.16
 Commit SHA: d2a013f
 Go version: go1.20.12
 OS/Arch: linux/amd64
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up
Generating certificates using cryptogen...
Creating docker network 'fabric_test'...
Starting peer0.org1.example.com...
Starting peer0.org2.example.com...
Starting orderer.example.com...
Fabric Network Started Successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ docker ps
CONTAINER ID   IMAGE                                COMMAND
CREATED         STATUS         NAMES
8a9f21b10c9d   hyperledger/fabric-peer:2.5.16     "peer node start"        1 minute
ago    Up 1 minute    peer0.org1.example.com
3c2b11e9a4f2   hyperledger/fabric-peer:2.5.16     "peer node start"        1 minute
ago    Up 1 minute    peer0.org2.example.com
1f9a88e22d3b   hyperledger/fabric-orderer:2.5.16 "orderer"                1 minute
ago    Up 1 minute    orderer.example.com
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh createChannel -c
mychannel
Generating genesis block for channel 'mychannel'...
Creating channel 'mychannel'...
Channel 'mychannel' created successfully
Joining peer0.org1 to channel 'mychannel'...
Successfully submitted proposal to join channel
Joining peer0.org2 to channel 'mychannel'...
Successfully submitted proposal to join channel
Setting anchor peer for Org1...
Anchor peer set for Org1
Setting anchor peer for Org2...
Anchor peer set for Org2
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Stopping peer0.org1.example.com ... done
Stopping peer0.org2.example.com ... done
Stopping  orderer.example.com   ...  done
Removing containers...
Removing network fabric_test
Network stopped successfully
```

RESULT

The Hyperledger Fabric development environment was successfully installed and configured. The required Fabric binaries and Docker images were installed, the test network was created successfully, the mychannel application channel was generated, both peer organizations joined the channel, and the anchor peers were configured successfully. The development environment is now ready for deploying chaincode (smart contracts) and developing blockchain applications.